Assignment Submission of Bike Rental Linear Regression

Upgrad & IIIT Banglore

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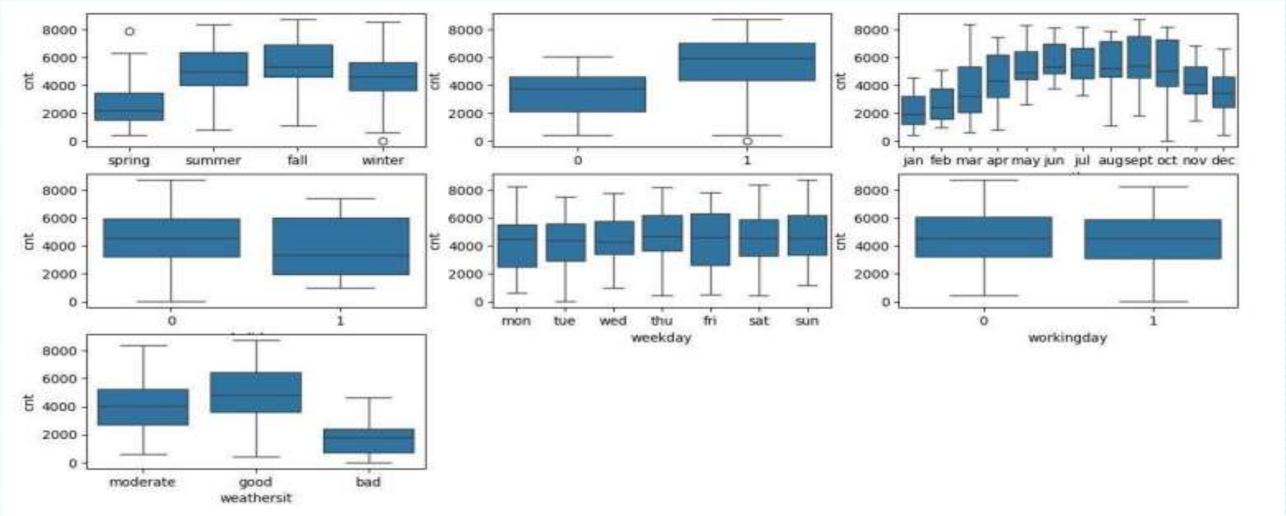
Assignment-Based Subjective Questions

01. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

Ans:- There are a couple of categorical variables namely season, month, year, weekday, working day and weathersit. These categorical variables have a major effect on the dependent variable ‘cnt’. The below fig shows the correlation among the same.



These variables are visualized using bar plot and Box plot both.

02. Why is it important to use drop_first=True during dummy variable creation?

Ans:- Dummy variable creation is a technique used in statistical modelling and machine learning to represent categorical variables with binary values (0 or 1). It involves creating new binary (dummy) variables for each category of the original categorical variable. These dummy variables serve as indicators for the presence or absence of a specific category. The number of dummy variables to create depends on the number of categories in the original categorical variable. For a categorical variable with n categories, you typically create $n - 1$ dummy variables.

Here's the rationale:

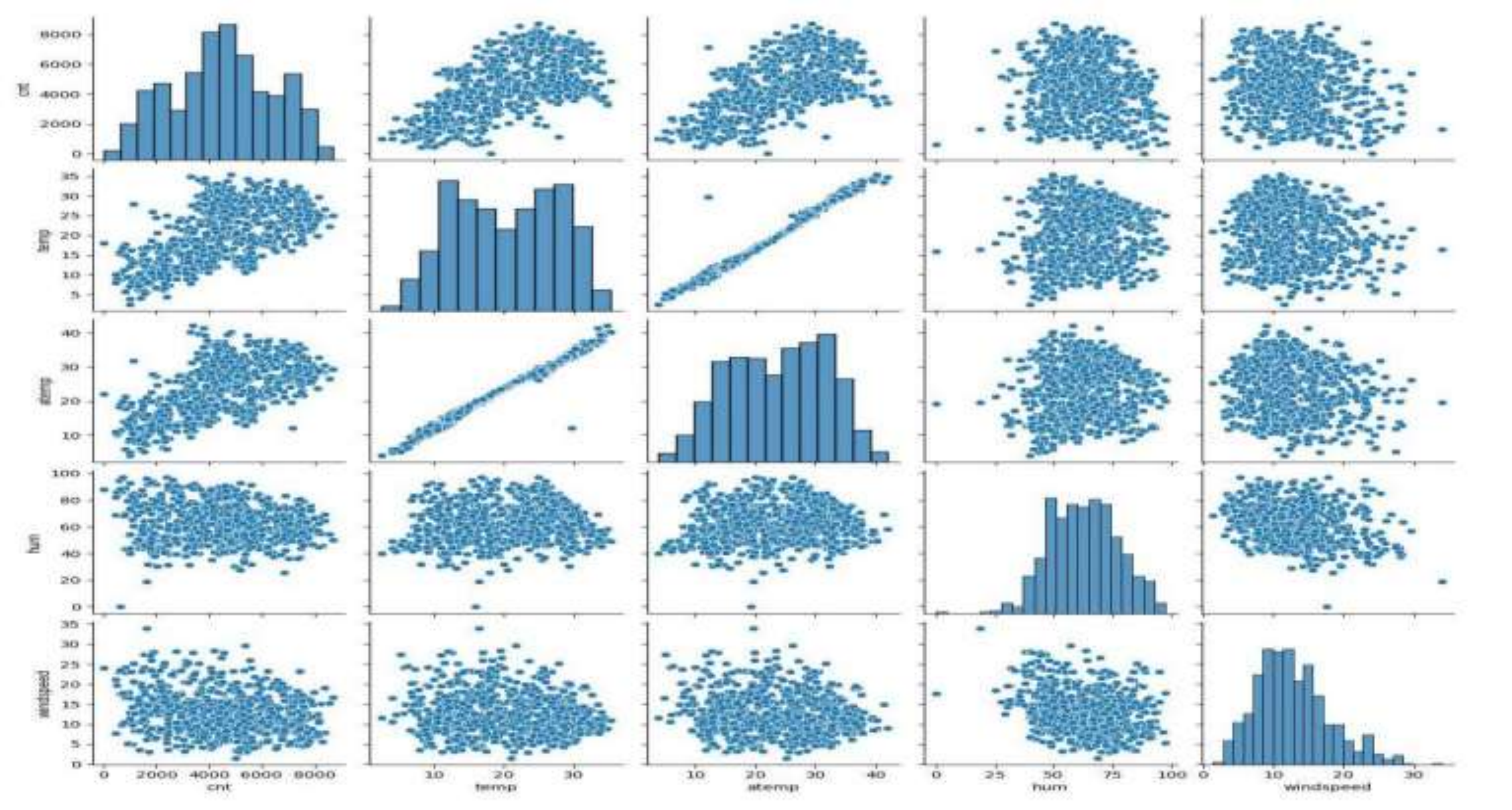
i. $\boldsymbol{n-1}$ Dummy Variables Rule: Creating n dummy variables introduces Multi-collinearity because the information in one category can be predicted from the others. This can lead to problems in the estimation of coefficients in regression models. By creating $n - 1$ dummy variables, you avoid perfect multi-collinearity, as the information about the omitted category is implicitly captured.

ii. Avoiding Redundancy: The information about the omitted category is implicitly captured by the constant term in the model. Including all dummy variables would introduce redundancy.

iii. Enhancing Interpretability: The coefficients of the dummy variables represent the change in the response variable compared to the omitted category. This makes the interpretation of the model more straightforward. For example, if you have a variable "Color" with categories "Red," "Blue," and "Green," you would create two dummy variables, say "Is_Blue" and "Is_Green." The absence of both dummy variables implies that the color is "Red." In Python, when using libraries like pandas, we can set `drop_first=True` while creating dummy variables to automatically drop one of the dummy variables to adhere to this $n - 1$ rule.

03.Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

Ans:- The 'temp' and 'atemp' variables have highest correlation when compared to the rest with target variable as 'cnt'.

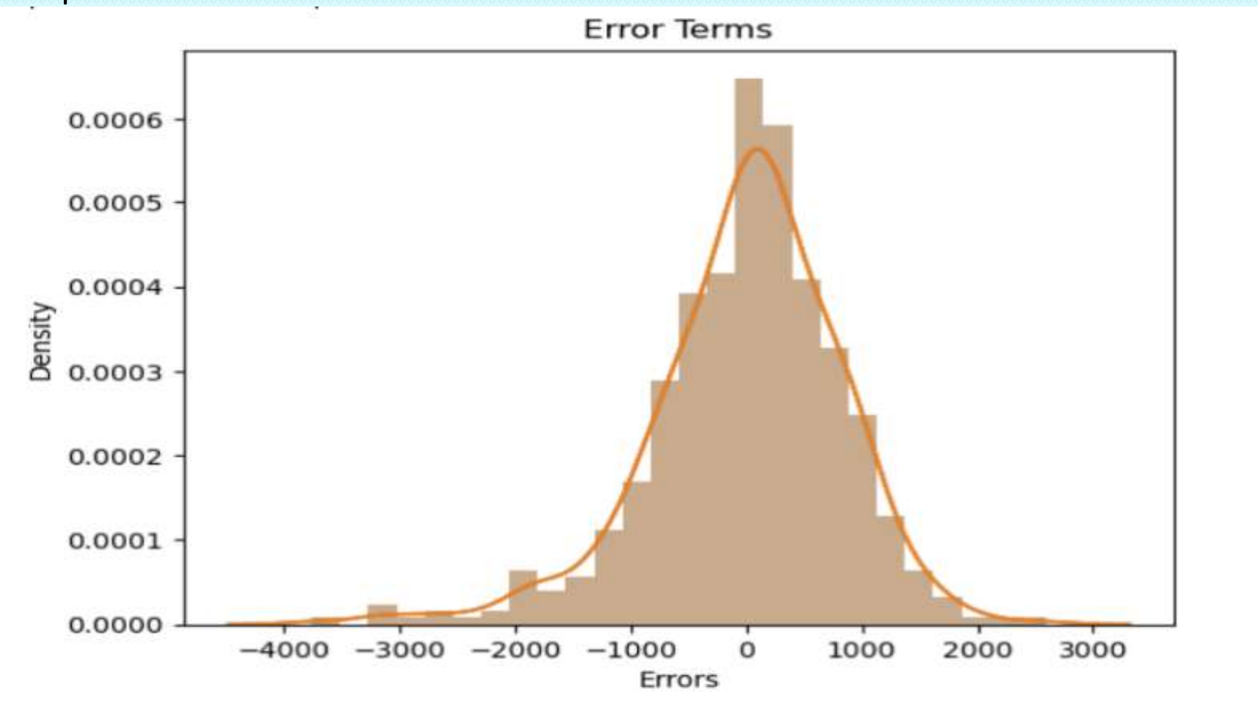


04.How did you validate the assumptions of Linear Regression after building the model on the training set?

Ans:-Validating the assumptions of linear regression is a crucial step to ensure the reliability of the model. After building the model on the training set, here are the steps I followed to validate the assumptions:

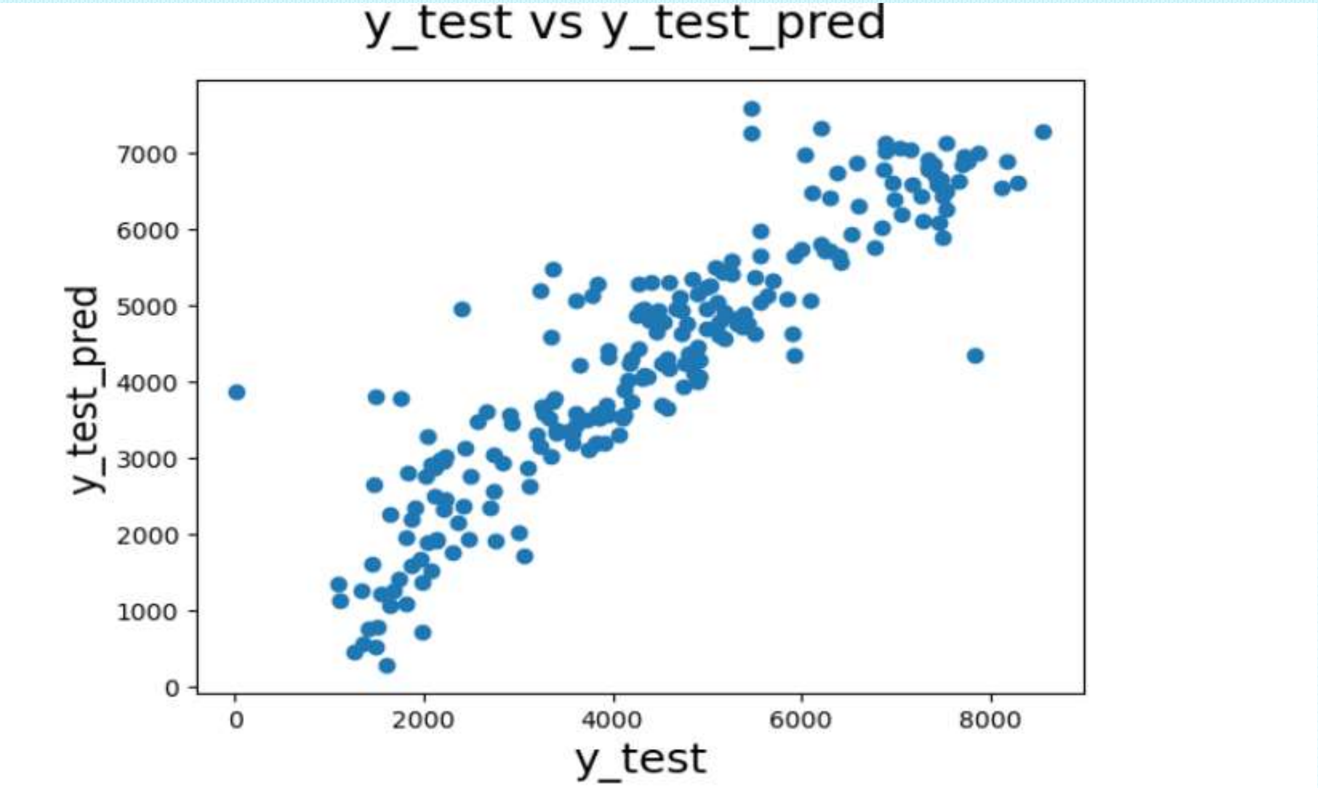
i. Residual Analysis:-

Process: Examine the residuals (the differences between observed and predicted values). - Check: Residuals should be approximately normally distributed, and there should be no discernible patterns in the residual plot.



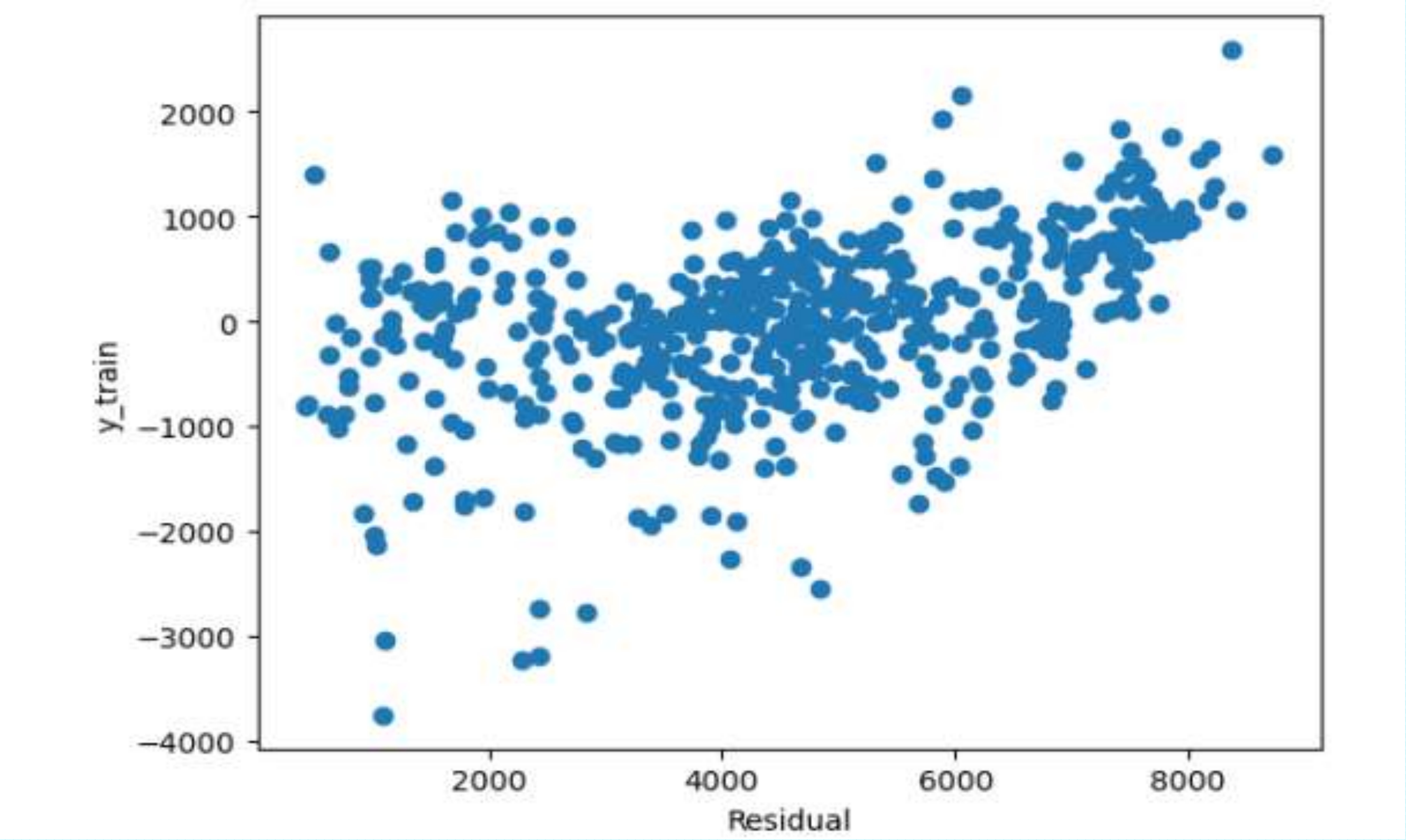
ii. Homoscedasticity (Constant Variance):

- Process: Plot residuals against predicted values.
- Check: The spread of residuals should be roughly constant across all levels of the predicted values.



iii. Linearity:

- Process: Create a scatterplot of observed vs. predicted values.
- Check: The points should fall approximately along a diagonal line, indicating a linear relationship.



iv. Independence of Residuals:

- Process: Examine residuals for autocorrelation.

- Check: There should be no discernible pattern in the residuals when plotted against >me or other relevant variables.

v. Multi-collinearity:

- Process: Calculate Variance Inflation Factors (VIF) for predictor variables.

- Check: VIF values should be below a certain threshold (commonly 5 or 10) to ensure no problematic multi-collinearity.

vi. Cross-Validation:

- Process: Validate the model on a test set or through cross-validation.

- Check: Assess the model's performance on new data to ensure generalizability and consistency.

vii. Check for Overfitting:

- Process: Evaluate model performance on a test set.

- Check: Ensure that the model generalizes well to new, unseen data without overfitting the training set.

05. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

Ans:- Top 3 features that has significant impact towards explaining the demand of the shared bikes are **temperature, year and season.**

General Subjective Questions

01. Explain the linear regression algorithm in detail?

Ans: Linear regression is a form of predictive modeling technique which tells us the relationship between the dependent (target variable) and independent variables (predictors). Since linear regression shows the linear relationship, which means it finds how the value of the dependent variable is changing according to the value of the independent variable.

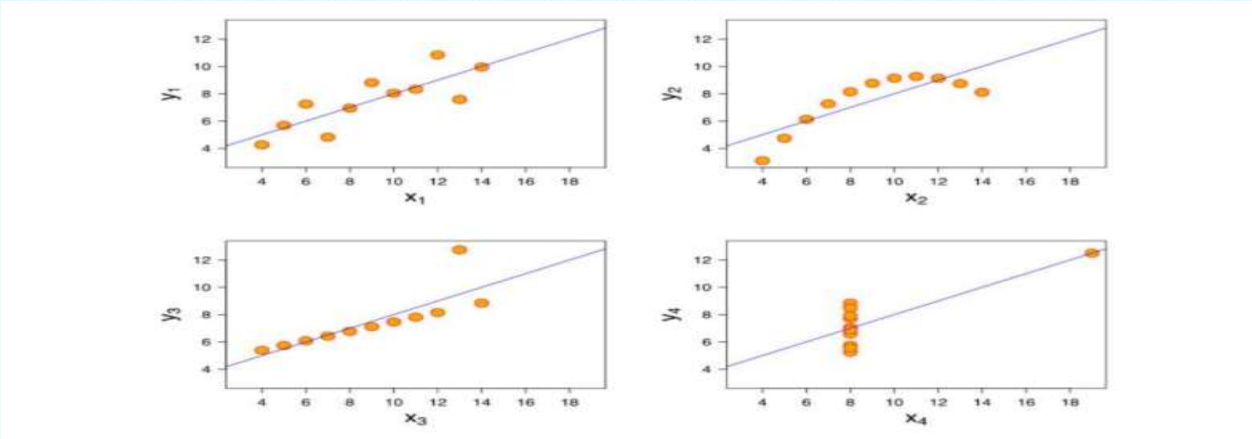
If there is a single input variable (x), such linear regression is called simple linear regression. And if there is more than one input variable, such linear regression is called multiple linear regression. The linear regression model gives a sloped straight line describing the relationship within the variables.

A regression line can be a Positive Linear Relationship or a Negative Linear Relationship. The goal of the linear regression algorithm is to get the best values for a0 and a1 to find the best fit line and the best fit line should have the least error.

In Linear Regression, RFE or Mean Squared Error (MSE) or cost function is used, which helps to figure out the best possible values for a0 and a1, which provides the best fit line for the data points.

02. Explain the Anscombe’s quartet in detail?

Ans: Anscombe’s Quartet can be defined as a group of four data sets which are nearly identical in simple descriptive statistics, but there are some peculiarities in the dataset that fools the regression model if built. They have very different distributions and appear differently when plotted on scatter plots. It was constructed to illustrate the importance of plotting the graphs before analyzing and model building, and the effect of other observations on statistical properties. There are these four data set plots which have nearly same statistical observations, which provides same statistical information that involves variance, and mean of all x, y points in all four datasets.



- 1st data set fits linear regression model as it seems to be linear relationship between X and y
- 2nd data set does not show a linear relationship between X and Y, which means it does not fit the linear regression model.
- 3rd data set shows some outliers present in the dataset which can’t be handled by a linear regression model.
- 4th data set has a high leverage point means it produces a high correlation coefficient. Its conclusion is that regression algorithms can be fooled so, it's important to data visualization before build machine learning model.

03. What is Pearson’s R?

Ans:- In Statistics, the Pearson's Correlation Coefficient is also referred to as Pearson's r, the Pearson product moment correlation coefficient (PPMCC), or bivariate correlation. It is a statistic that measures the linear correlation between two variables.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Here

- X% and Y% are the individual data points.

- X ^ and Y ^ are the means of X and Y.

- The numerator represents the covariance between X and Y.

- The denominator is the product of the standard deviations of X and Y.

- Pearson's correlation coefficient is widely used in statistics to assess the strength and direction of the linear relationship between two variables. It's important to note that correlation does not imply causa>on, and a correlation coefficient close to zero does not necessarily mean the absence of a relationship; it only indicates the absence of a linear relationship.

04. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Ans: Scaling means you're transforming your data so that it fits within a specific scale. It is one type of data preprocessing step where we will fit data in specific scale and speed up the calculations in an algorithm.

Collected data contains features varying in magnitudes, units and range. If scaling is not performed than algorithm tends to weigh high values magnitudes and ignore other parameters which will result in incorrect modeling.

Difference between Normalizing Scaling and Standardize Scaling:

i. In normalized scaling minimum and maximum value of features being used whereas in Standardize scaling mean and standard deviation is used for scaling.

ii. Normalized scaling is used when features are of different scales whereas standardized scaling is used to ensure zero mean and unit standard deviation.

iii. Normalized scaling scales values between (0,1) or (-1,1) whereas standardized scaling is not having or is not bounded in a certain range.

iv. Normalized scaling is affected by outliers whereas standardized scaling is not having any effect by outliers.

v. Normalized scaling is used when we don't know about the distribution whereas standardized scaling is used when distribution is normal.

05. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

Ans: VIF (Variance Inflation Factor) basically helps explain the relationship of one independent variable with all the other independent variables. The formulation of VIF is given below:

A VIF value of greater than 10 is definitely high, a VIF of greater than 5 should also not be ignored and inspected appropriately. A very high VIF value shows a perfect correlation between two independent variables. In the case of perfect correlation, we get R2 =1, which lead to 1/(1-R2) infinity. To solve this problem we need to drop one of the variables from the dataset which is causing this perfect multi collinearity.

06. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

Ans: Q–Q plot is a probability plot, which is a graphical method for comparing two probability distributions by plotting their quantiles against each other.

Quantile-Quantile (Q-Q) plot, is a graphical tool to help us assess if a set of data possibly came from some theoretical distribution such as a Normal, exponential or Uniform distribution. QQ plot can also be used to determine whether or not two distributions are similar or not. If they are quite similar you can expect the QQ plot to be more linear. The linearity assumption can best be tested with scatter plots. Secondly, the linear regression analysis requires all variables to be multivariate normal. This assumption can best be checked with a histogram or a Q-Q-Plot

Importance of QQ Plot in Linear Regression: In Linear Regression when we have a train and test dataset then we can create Q-Q plot by which we can confirm that both the data train and test data set are from the population with the same distribution or not.

Advantages:

- It can be used with sample size also
- Many distributional aspects like shifts in location, shifts in scale, changes in symmetry, and the presence of outliers can all be detected from this plot
- Q-Q plot use on two datasets to check
- If both datasets came from population with common distribution
- If both datasets have common location and common scale
- If both datasets have similar type of distribution shape
- If both datasets have tail behavior